

Multilingual NLP and Cross-lingual Transfer

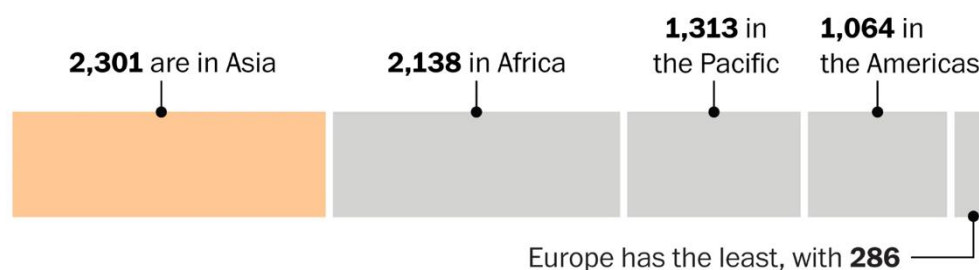
David Ifeoluwa Adelani

COMP-550

March 30, 2026

Why Research on Multilingual NLP?

- There are **7151 languages** in the world (Ethnologue 2022)
- Languages are **not treated equally** by researchers
 - Number of NLP publications extracted from ACL Anthology
 - Fewer representation of **African**, **Asian** and **Americas** languages



Number of NLP publications per language



How many languages are supported?

Highly debatable? 200, 500, 1000, 1600?

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Scaling neural machine t

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Building Machine Translation Systems for the Next Thousand Languages

Ankur Bapna · [Isaac Caswell](#) · Julia Kreutzer · [Orhan Firat](#) · [Daan van Esch](#) · Aditya Siddhant · Mengmeng Niu · Pallavi Nikhil Baljekar · Xavier Garcia · [Wolfgang Macherey](#) · [Theresa Breiner](#) · Vera Saldinger Axelrod · [Jason Riesa](#) · Yuan Cao · [Mia Chen](#) · [Klaus Macherey](#) · Maxim Krikun · Pidong Wang · [Alexander Gutkin](#) · Apu Shah · Yanping Huang · Zhifeng Chen · Yonghui Wu · Macduff Richard Hughes · Google Research (2022)

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Glott500:

Scaling Multilingual Corpora and Language Models to 500 Languages

Ayyoob Imani^{*1,2}, Peiqin Lin^{*1,2}, Amir Hossein Kargaran^{1,2}, Silvia Severini¹,

Ma^{1,2},

Hinrich Schütze^{1,2}

g (MCML), Germany

telecomunicações

nce

Llama

However, many are not supported in Evaluation?

Often default to Bible-based benchmark e.g Taxi1500

Omnilingual MT: Machine Translation for 1,600 Languages

March 17, 2026

Under-resourced languages: Labelled+Unlabelled data

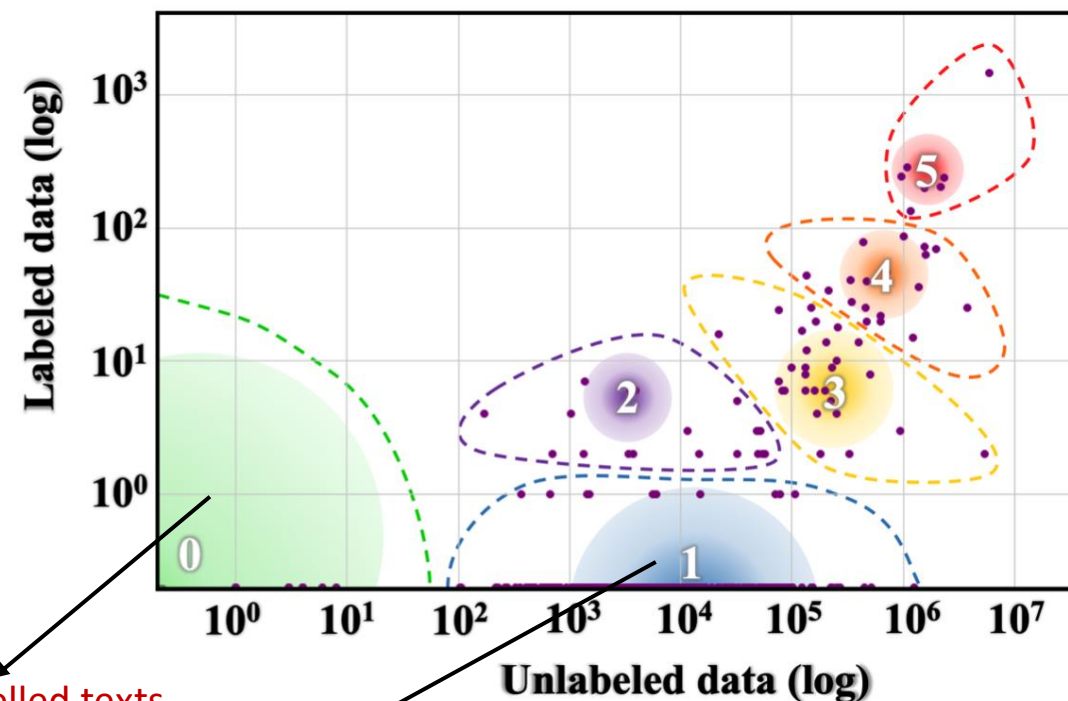
- **Joshi et al (2020)** proposed a **six-class categorization** of languages based on
 - Unlabelled corpora
 - Labelled corpora

Class	5 Example Languages	#Langs
0	Dahalo, Warlpiri, Popoloca, Wallisian, Bora	2191
1	Cherokee, Fijian, Greenlandic, Bhojpuri, Navajo	222
2	Zulu, Konkani, Lao, Maltese, Irish	19
3	Indonesian, Ukranian, Cebuano, Afrikaans, Hebrew	28
4	Russian, Hungarian, Vietnamese, Dutch, Korean	18
5	English, Spanish, German, Japanese, French	7

Which other language is in class 5 (Winners)?

class 4 (Under-dogs)?

class 0 (Left behind)?



No unlabelled texts
80% of languages

Few texts

NLP in a Nutshell (2018 – Today)

Pre-train -> **Fine-tune**

sentiment
classification,
QA ...

LSTM or
MLP

Fine-tuning

Word2Vec
BERT
T5

E.g. Masked LM on
unlabelled texts

Self-supervised **pre-training**

Pre-train -> **Prompt** -> **Predict**

sentiment
classification,
QA ...

Predict with a **verbalizer**

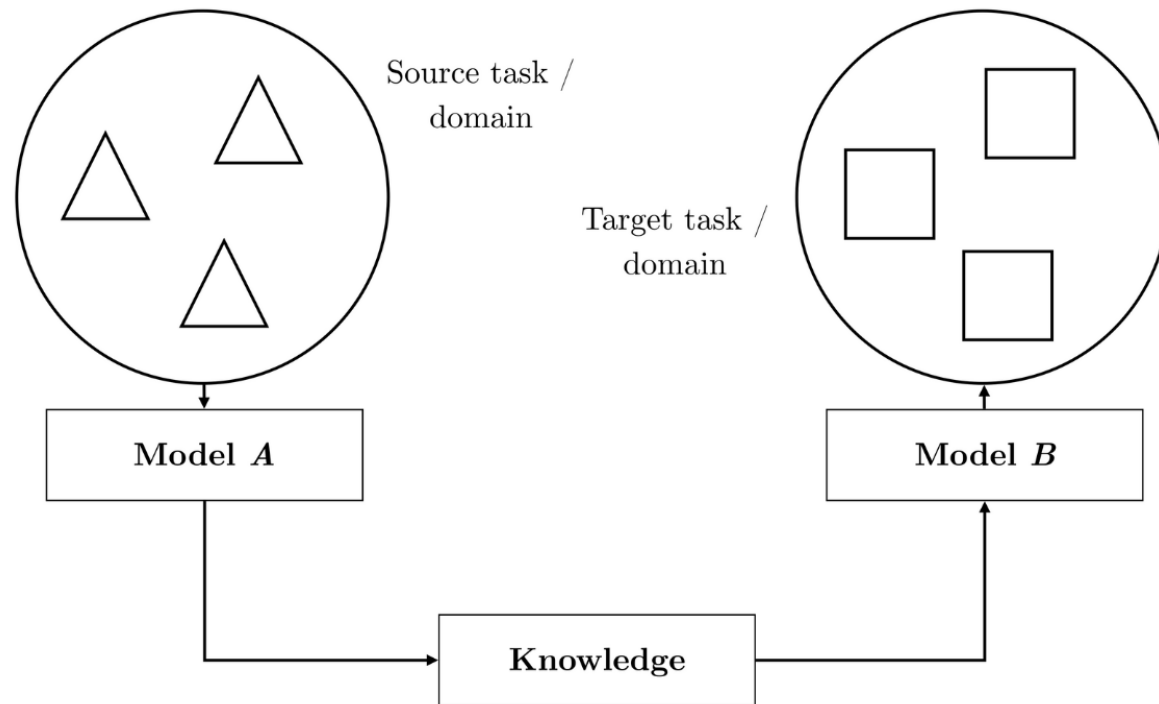
Prompt with or
without examples

E.g. Causal LM on
unlabelled texts

Self-supervised **pre-training**
Or **next-word prediction**

Cross-lingual transfer

What is transfer learning?



- Inspired by transfer learning of **ImageNet** to **other computer vision tasks**
- **Same task: labelled data** in a **source domain** transferred to **target domains**.
 - Same for **source language** to **target language** transfer
- **Different tasks: labelled data** in the **target domains**.
 - Sequential learning like BERT -> sentiment classification
 - Multi-task learning

Monolingual transfer learning

Pre-train -> **Fine-tune**

sentiment
classification,
QA ...

LSTM or
MLP

Fine-tuning

Word2Vec
BERT
T5

E.g. Masked LM on
unlabelled texts

Self-supervised **pre-training**

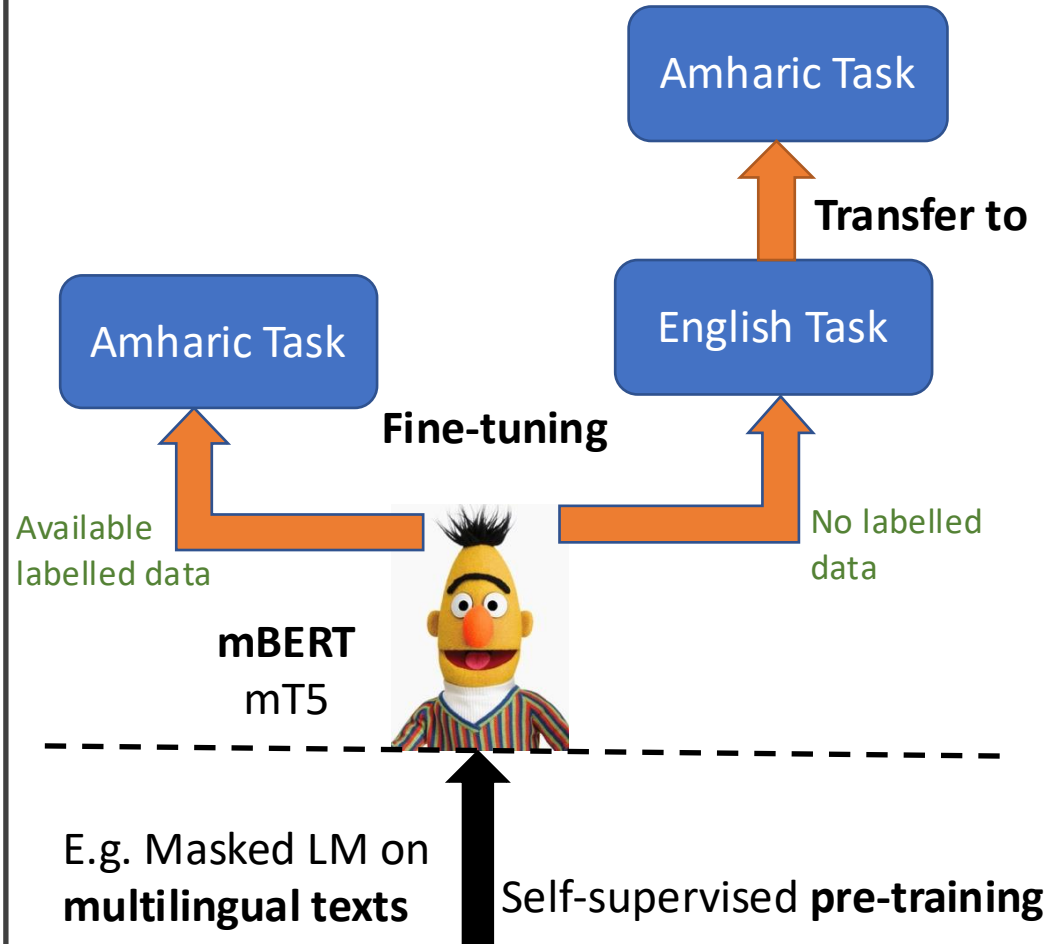
- **Feature extraction**

- Extract sentence embeddings for LSTM / CNN classifier
- Weights are frozen

- **Fine-tuning**

- End-to-end fine-tuning
- All weights are modified
- Often leads to better performance

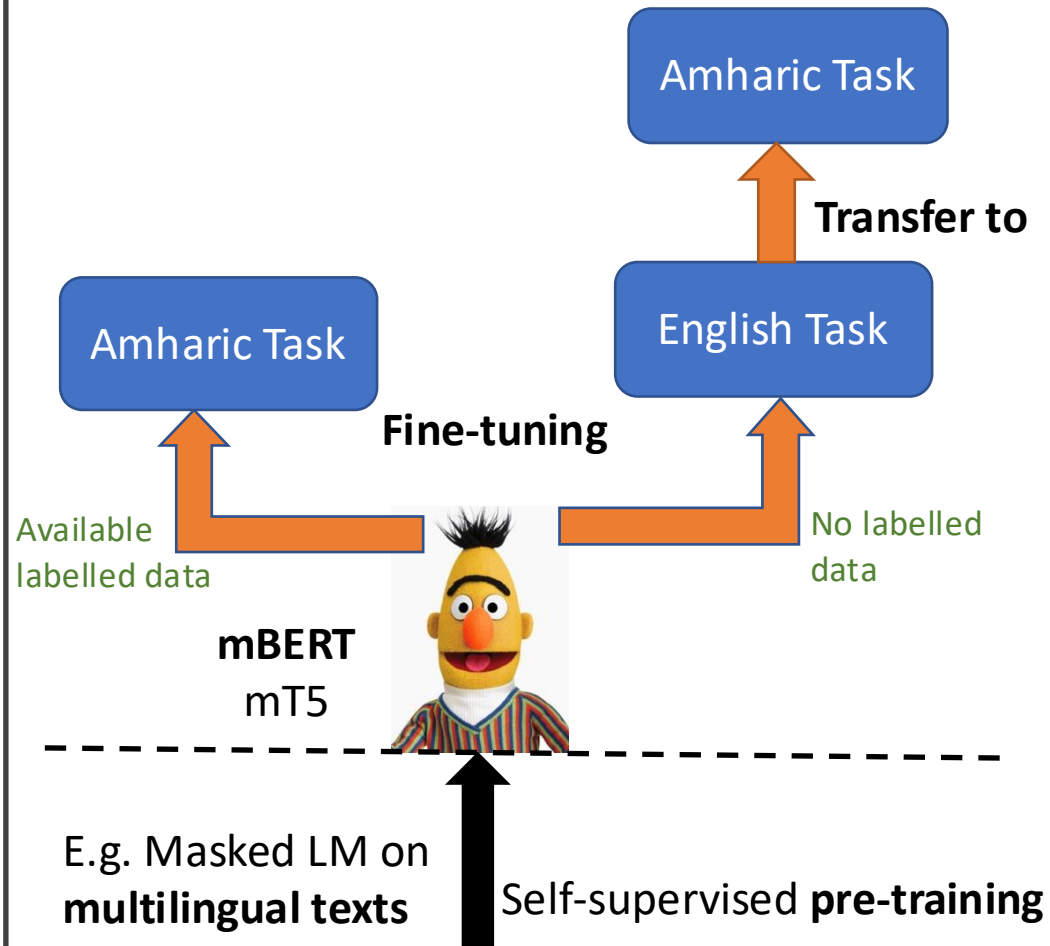
Cross-lingual Transfer learning



We leverage a multilingual PLM instead of a monolingual one

Suppose we want to apply **transfer learning to a new language** not covered in the pre-training of mBERT e.g. **Amharic**

Cross-lingual Transfer learning



Most pre-trained LMs only **cover 100 languages**, and many are not even evaluated on downstream tasks

<=60
Languages

VECO, BLOOM, mBART, mGPT, Gemini, GPT-4

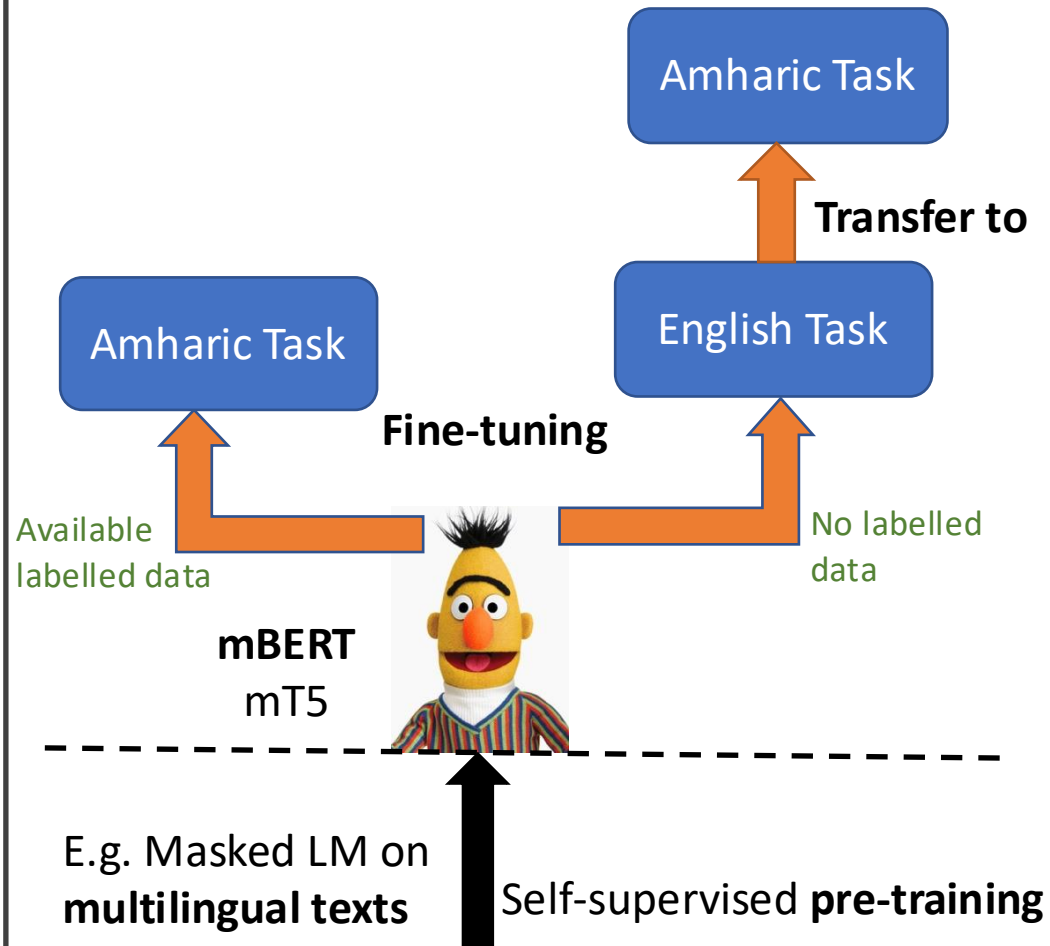
100-120
Languages

mBERT, XLM-R, RemBERT, InfoXLM, mT5, XGLM

200-500
Languages

NLLB-200, Glot500, Serengeti

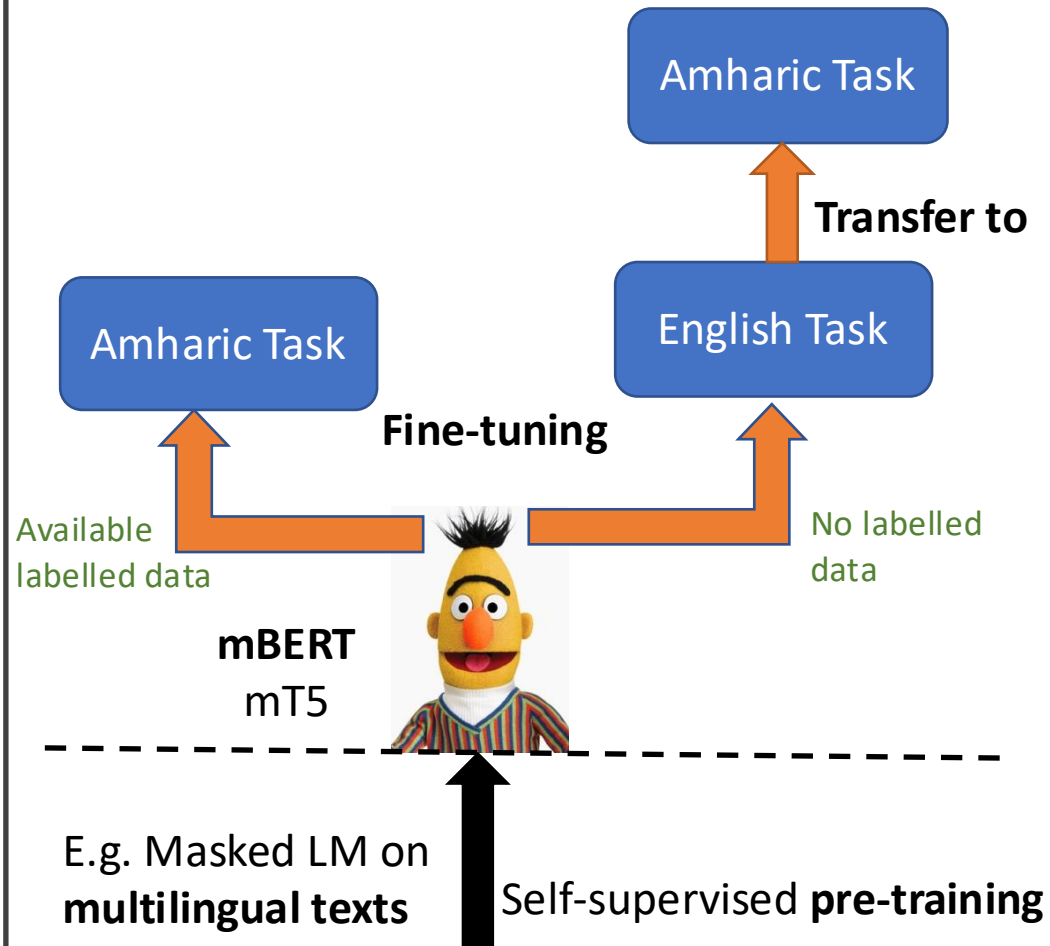
Cross-lingual Transfer learning



Cross-lingual transfer learning is **attractive** because there are over **7,000+** languages, many do **lack labelled data**



Cross-lingual Transfer learning



Challenges:

- Lack of **labelled datasets**
- Most PLMs only **cover few languages** (up to. 100)
- Source languages and target languages may use **different scripts**.
- Multiple fine-tuning leads to **catastrophic forgetting** and **parameter inefficient** (multiple checkpoints per fine-tuning runs)
- What is the **best source language** to transfer from?

Challenge 1: Benchmark datasets

Benchmark and leaderboard powers progress

- WikiANN (Pan et al., 2017)
- XNLI (Conneau et al., 2018)
- XQUAD (Artetxe et al., 2020)
- **XTREME (Hu et al., 2020)**

XNLI: Evaluating Cross-lingual Sentence Representations

Alexis Conneau, Guillaume Lample, Ruty Rinott, Holger Schwenk, Ves Stoyanov
Facebook AI
aconneau, glample, ruty, schwenk, ves@fb.com

Adina Williams, Samuel R. Bowman
New York University
adinawilliams, bowman@nyu.edu

Cross-lingual Name Tagging and Linking for 282 Languages

Xiaoman Pan¹, Boliang Zhang¹, Jonathan May²,
Joel Nothman³, Kevin Knight², Heng Ji¹

On the Cross-lingual Transferability of Monolingual Representations

Mikel Artetxe^{†*}, Sebastian Ruder[‡], Dani Yogatama[‡]

XTREME: A Massively Multilingual Multi-task Benchmark for Evaluating Cross-lingual Generalization

Junjie Hu^{*1} Sebastian Ruder^{*2} Aditya Siddhant³ Graham Neubig¹ Orhan Firat³ Melvin Johnson³

Task	Corpus	Train	Dev	Test	Test sets	Lang.	Task	Metric	Domain
Classification	XNLI	392,702	2,490	5,010	translations	15	NLI	Acc.	Misc.
	PAWS-X	49,401	2,000	2,000	translations	7	Paraphrase	Acc.	Wiki / Quora
Struct. pred.	POS	21,253	3,974	47-20,436	ind. annot.	33 (90)	POS	F1	Misc.
	NER	20,000	10,000	1,000-10,000	ind. annot.	40 (176)	NER	F1	Wikipedia
QA	XQuAD	87,599	34,726	1,190	translations	11	Span extraction	F1 / EM	Wikipedia
	MLQA			4,517-11,590	translations	7	Span extraction	F1 / EM	Wikipedia
	TyDiQA-GoldP	3,696	634	323-2,719	ind. annot.	9	Span extraction	F1 / EM	Wikipedia
Retrieval	BUCC	-	-	1,896-14,330	-	5	Sent. retrieval	F1	Wiki / news
	Tatoeba	-	-	1,000	-	33 (122)	Sent. retrieval	Acc.	misc.

Model	Participant	Affiliation	Submission Date	Score	Sentence-pair Classification	Structured Prediction	Question Answering	Sentence Retrieval	Number of parameters (in millions)
	Human	-	-	93.3	95.1	97.0	87.8	-	
VECO 2.0	AliceMind	Alibaba	March 17, 2023	85.8	90.8	84.6	77.2	94.9	
Turing ULR v6	Alexander v-team	Microsoft	Sep 6, 2022	85.5	91.0	83.8	77.1	94.4	
ShenNonG	Cloud Xiaowei AI	Tencent	May 22, 2022	85.0	90.4	83.1	76.3	94.4	
Turing ULR v5	Alexander v-team	Microsoft	Nov 24, 2021	84.5	90.3	81.7	76.3	93.7	

Challenge 1: Grassroots Benchmark efforts

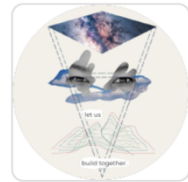
- Most **multilingual benchmark** are more **western-centric** or focus on **resource-rich regional languages** e.g. Hindi, Bengali, Indonesia, Swahili.
- In early 2020, different grassroots movement began to develop novel datasets for regional languages e.g.
 - India
 - Africa
 - Indonesia & South-East Asia
 - Indigeneous Americas languages



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🔗 <https://groups.google.com/u/0/g/se...> ✉ seacrowd.research@gmail.com

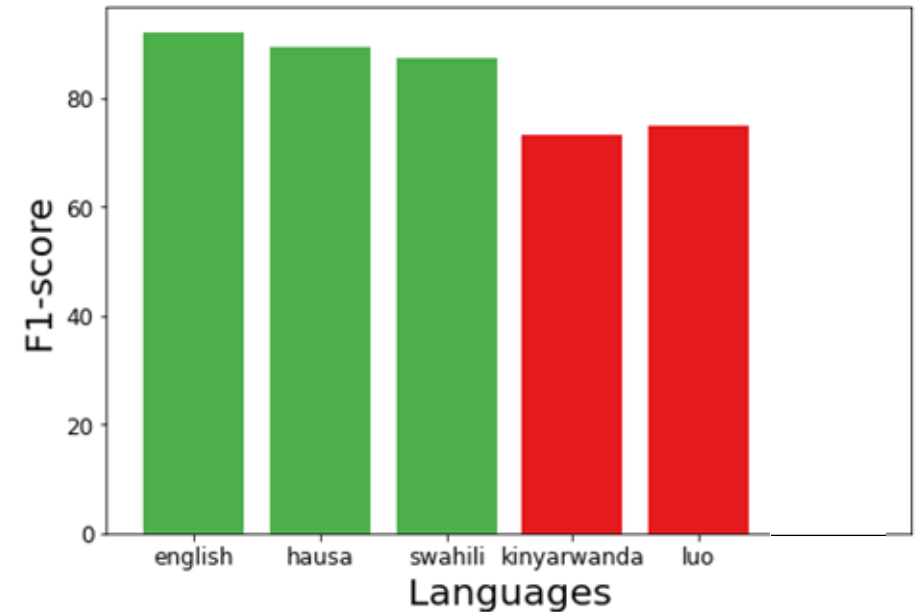
Challenge 2: Limited coverage in LMs

Limited language coverage in PLMs

- **Curse of multilinguality**
 - **Non-availability of corpora**
 - Most PLMs limits coverage to 100 languages
- **Significant performance drop on unseen languages**

Addressing Cross-lingual transfer learning

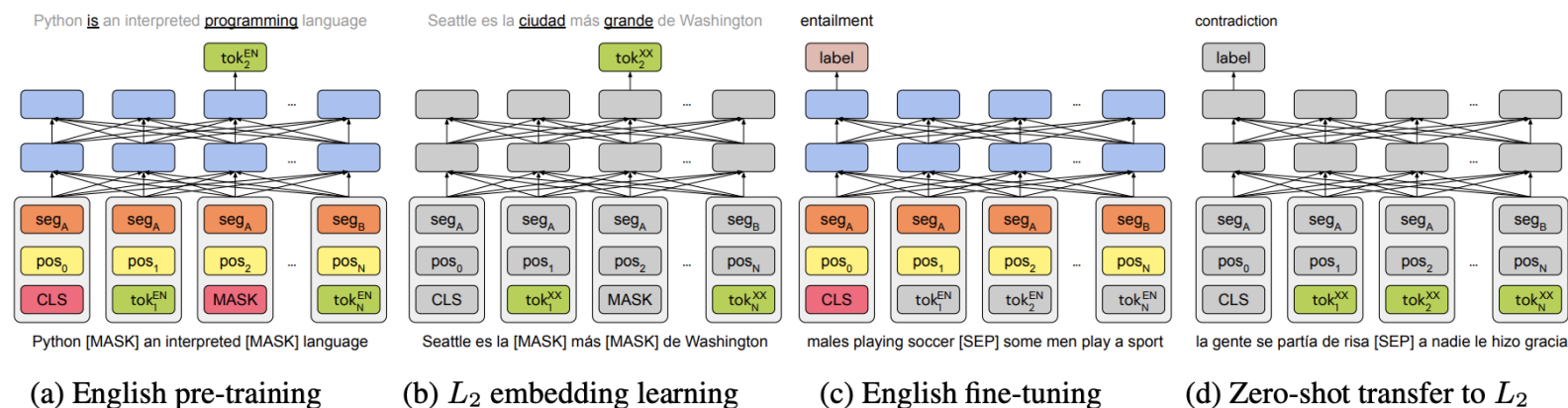
1. Monolingual representations adaptation
2. Monolingual Adaptation with active forgetting
3. Language adaptive pre-training/fine-tuning
4. Multilingual adaptive fine-tuning
5. Parameter-efficient adaptation approaches
6. Continue pre-training + Supervised fine-tuning
7. LangBridge and MindMerger



Performance of PLM on MasakhaNER data
Kinyarwanda, and luo were not used in pretraining

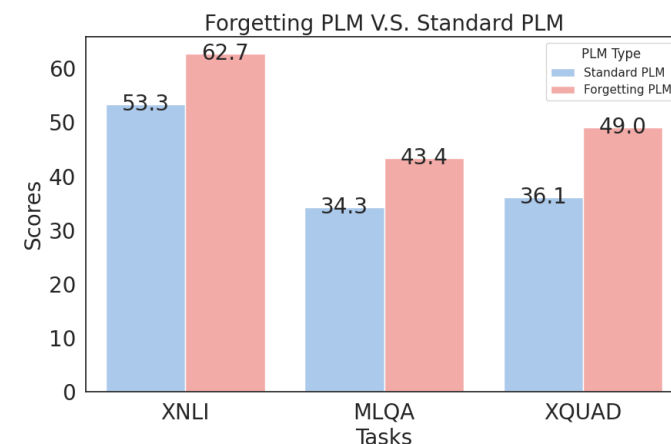
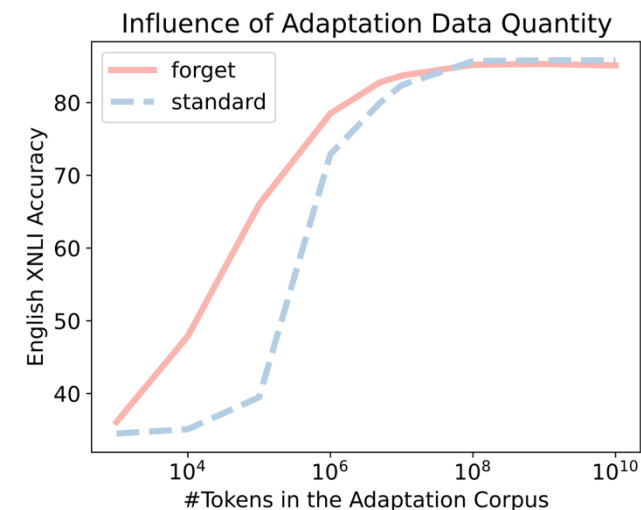
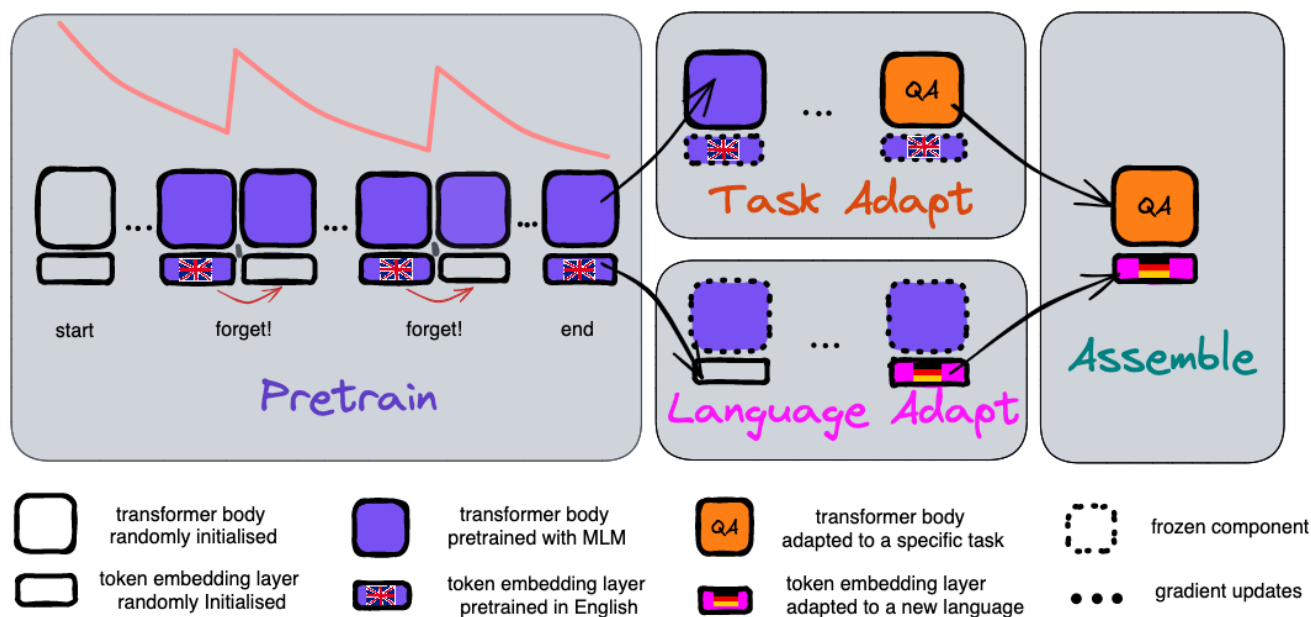
Monolingual representations adaptation

- Adaptation with a monolingual BERT to other languages
- Following four steps:
 - a) **Pre-train a BERT model** in English
 - b) **Freeze the transformer body** and **learn new token embeddings from scratch** for a second language using monolingual corpus and MLM objective
 - c) **Fine-tune the model on English** while keeping the **embeddings frozen**
 - d) **Zero-shot transfer** it to the **new language by swapping the token embeddings**.



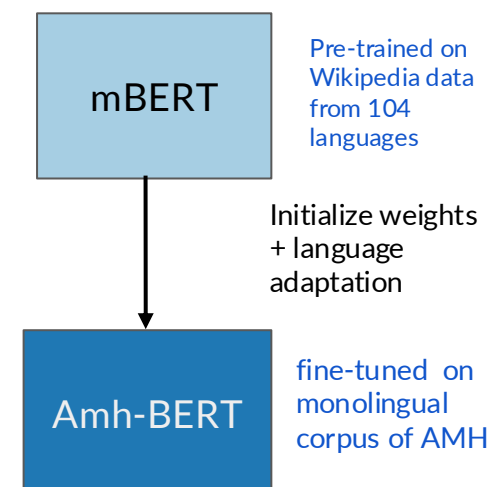
Monolingual adaptation with Active forgetting

- Same as adaptation with a monolingual BERT to other languages
- **Active forgetting**: resets token embeddings at regular intervals, while leaving all other parameters untouched throughout pretraining.
 - Makes adaptation to distant languages easier, because pre-training is more robust to noise



Language adaptive fine-tuning

- Perform *language adaptation* using multilingual PLM
 - **Language Adaptive Pre-training/Fine-tuning (LAPT/LAFT)**
 - Fine-tune a PLM on monolingual corpus using the same pre-training objective
- **Vocabulary augmentation** for new languages and scripts
 - 99 unused wordpieces in mBERT (Chau & Smith, 2021)
 - Vocabulary replacement is also possible (Alabi et al., 2020)



Rep.	BE* (0)	BG (0)	GA (1)	MT (2)	UG (2)	UR (0)	VI (0)	WO (2)	Avg.
FASTT	68.84 ± 7.16	88.86 ± 0.37	86.87 ± 2.55	89.68 ± 2.15	89.45 ± 1.37	90.81 ± 0.31	81.84 ± 1.15	87.48 ± 0.55	85.48
BERT	91.00 ± 0.30	94.48 ± 0.10	90.36 ± 0.20	92.61 ± 0.10	90.87 ± 0.13	89.88 ± 0.13	84.73 ± 0.13	87.71 ± 0.31	90.20
mBERT	94.57 ± 0.45	96.98 ± 0.08	91.91 ± 0.25	94.01 ± 0.17	78.07 ± 0.22	91.77 ± 0.18	88.97 ± 0.10	93.04 ± 0.20	91.16
LAPT	95.74 ± 0.44	97.15 ± 0.04	93.28 ± 0.19	95.76 ± 0.09	79.88 ± 0.27	92.18 ± 0.16	89.64 ± 0.20	94.58 ± 0.13	92.28
VA	95.28 ± 0.51	97.20 ± 0.06	93.33 ± 0.16	96.33 ± 0.09	91.49 ± 0.13	92.24 ± 0.16	89.49 ± 0.22	94.48 ± 0.20	93.73

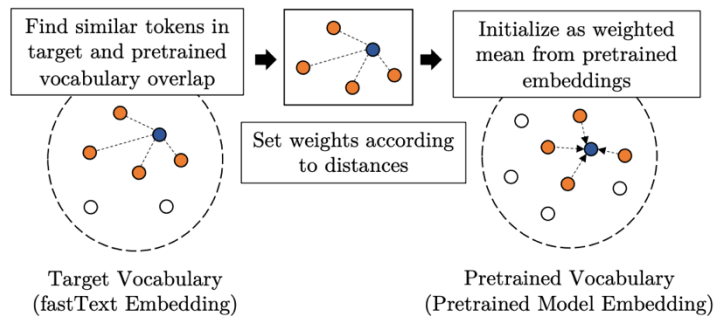
(a) POS tagging (accuracy). *Belarusian uses universal POS tags.

Alabi J. et al. 2020. Massive vs. Curated Embeddings for Low-Resourced Languages: the Case of Yorùbá and Twi. In LREC 2020.

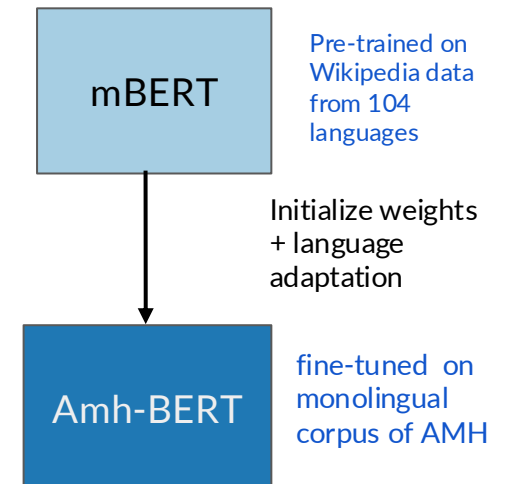
Chau Ethan C and Smith, Noah. Specializing Multilingual Language Models: An Empirical Study. In MRL @EMNLP2021

Challenge 3: Adaptation to new scripts

- **Vocabulary replacement** can be effective for adapting to **new scripts**.
 - **Random initialization** of added sub-word tokens can be improved.
 - WECHSEL
 - **FOCUS** – leverages external word embeddings like FastText
 - OFA – supports more than one language



Method	XNLI (translate-train)			
	German	Arabic	Kiswahili	Avg.
XLM-R (original vocab)				
- off-the-shelf	78.8 ± 0.3	74.8 ± 0.4	69.1 ± 0.4	74.2
- LAPT	78.9 ± 0.4	75.1 ± 0.6	72.4 ± 0.4	75.5
XLM-R (replaced vocab)				
- Random + LAPT [†]	77.6 ± 0.4	74.6 ± 0.4	71.2 ± 0.3	74.5
- WECHSEL _{EN} + LAPT	77.7 ± 0.5	75.4 ± 0.3	72.0 ± 0.2	75.0
- WECHSEL + LAPT	78.2 ± 0.2	76.0 ± 0.2	72.3 ± 0.3	75.5
- FOCUS + LAPT	78.3 ± 0.6	76.5 ± 0.4	72.9 ± 0.5	75.9
XLM-R (extended vocab)				
- Random + LAPT	77.7 ± 0.6	75.2 ± 0.6	71.8 ± 0.4	74.9
- FOCUS + LAPT	78.0 ± 0.4	75.5 ± 0.4	72.1 ± 0.2	75.2



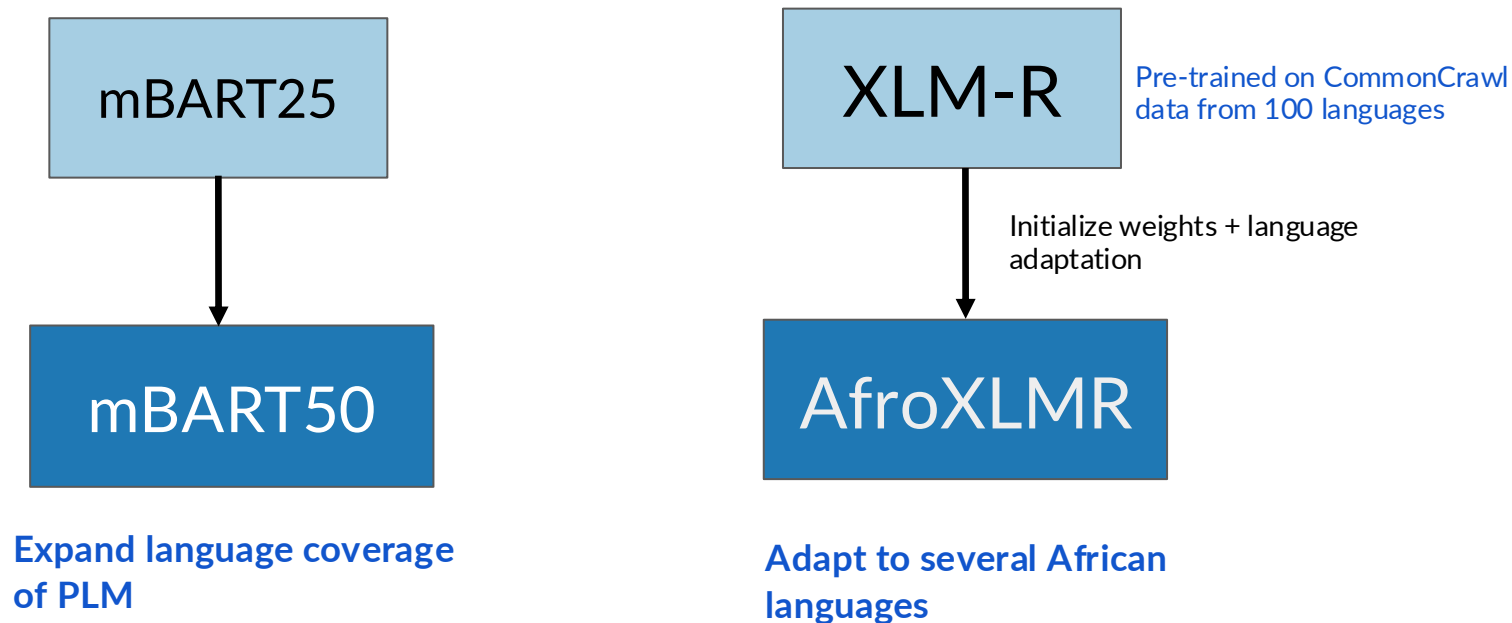
Limitations of LAFT

Some limitations

- **Difficult to adapt for non-supported scripts**
- **LAFT creates a copy** of the PLM model for each language
 - **parameter inefficient** for many languages.
- **Limit the cross-lingual transfer abilities** of the adapted model

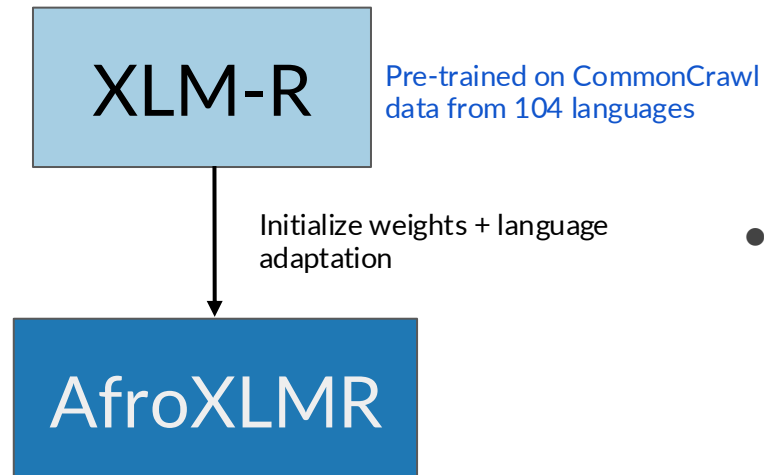
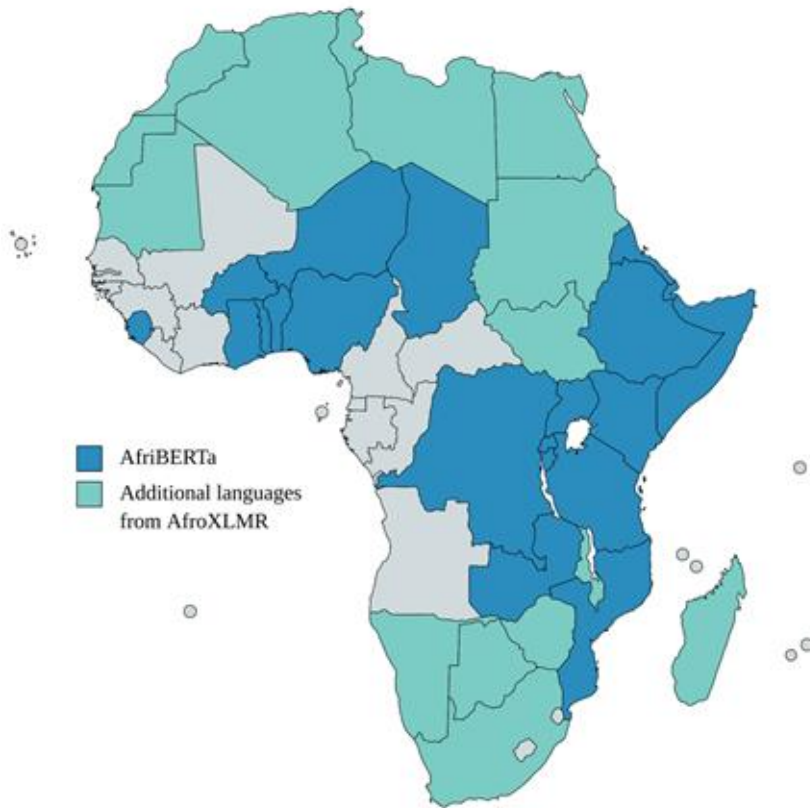
Multilingual Adaptive Fine-tuning (MAFT)

MAFT To address limited language coverage



Tang et al. **Multilingual translation with extensible multilingual pretraining and finetuning.** arXiv 2020

MAFT for African languages



- **Choice of PLM**

- **XLM-R-base**
- Covers **Arabic, Latin** and **Ge'ez** scripts

- **Languages:**

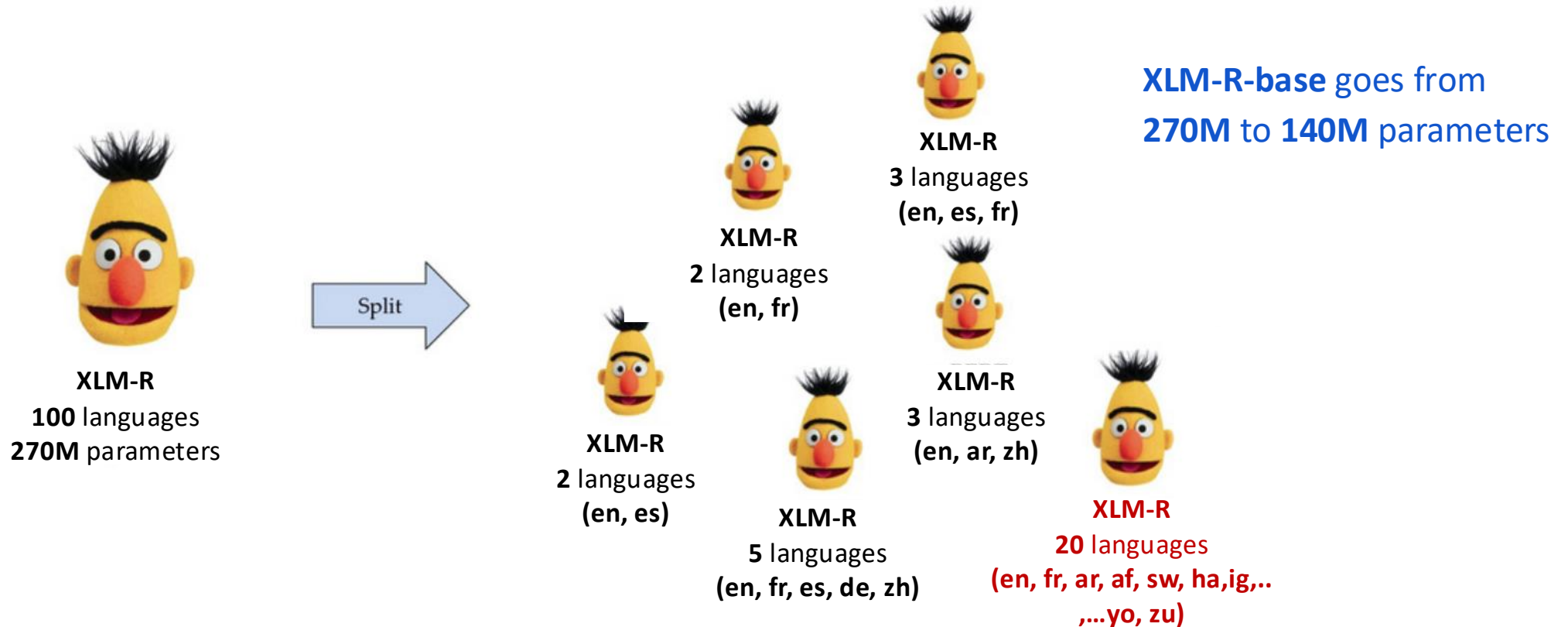
- **English, French, Arabic,** and
- **17 most-resourced** African languages (>50MB)

- **Corpus**

- mC4 – filtered web archive
- News corpus

Vocabulary reduction before MAFT

- An approach that leads to **competitive performance** compared to the **original multilingual PLM**
- **Vocab reduction + MAFT**



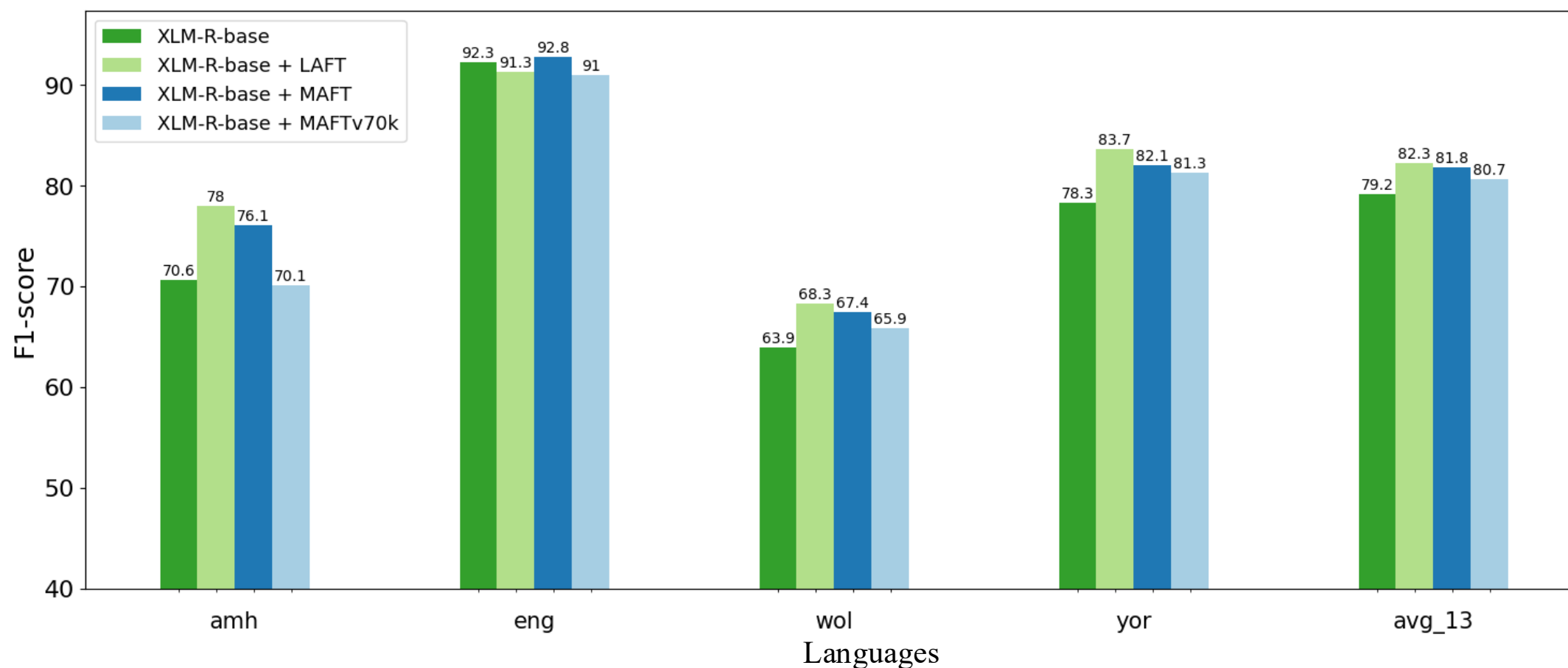
Evaluation on Named Entity Recognition (NER)

- A core task in information extraction
- Recognizing entities like
 - **Person**,
 - **Location**,
 - **Organization** or
 - **Date**.
- **Token** classification task
- Evaluation is performed using F1-score

On	O
the	O
4th	B-DATE
of	I-DATE
February	I-DATE
,	O
Global	B-ORG
Voices	I-ORG
visited	O
Fernando	B-PER
Gomes	I-PER

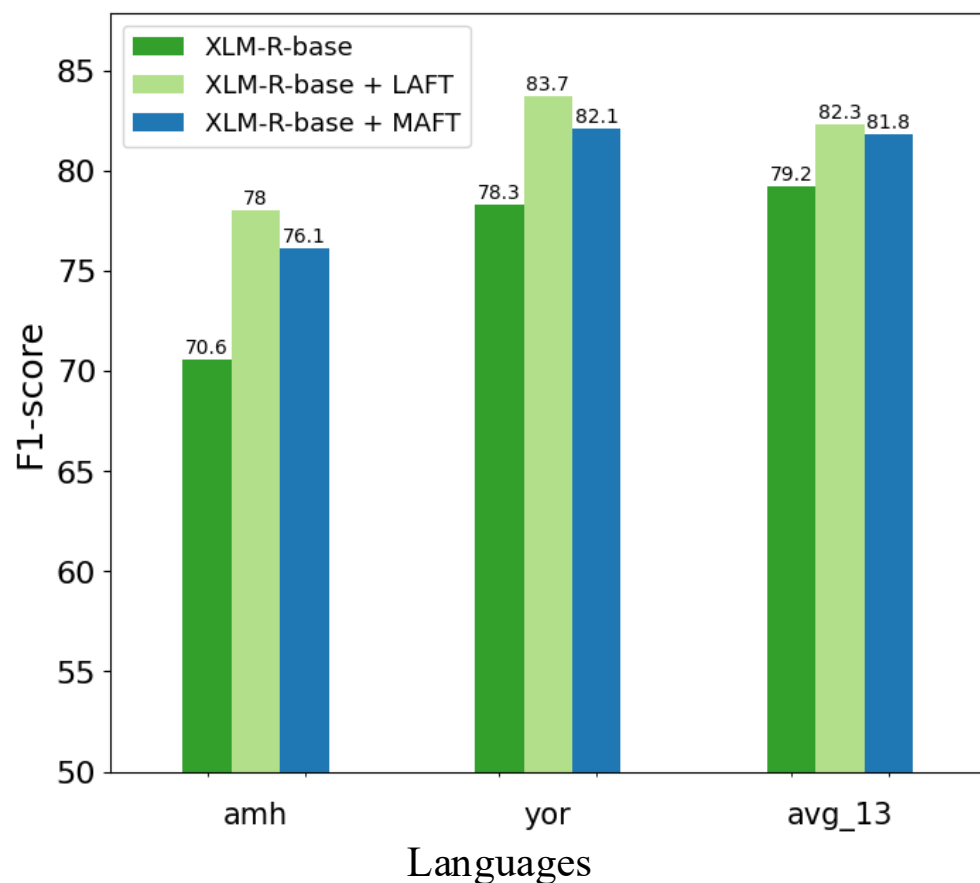
Evaluation on MAFT on NER and Vocab compression

Vocab reduction leads to
slight drop in performance

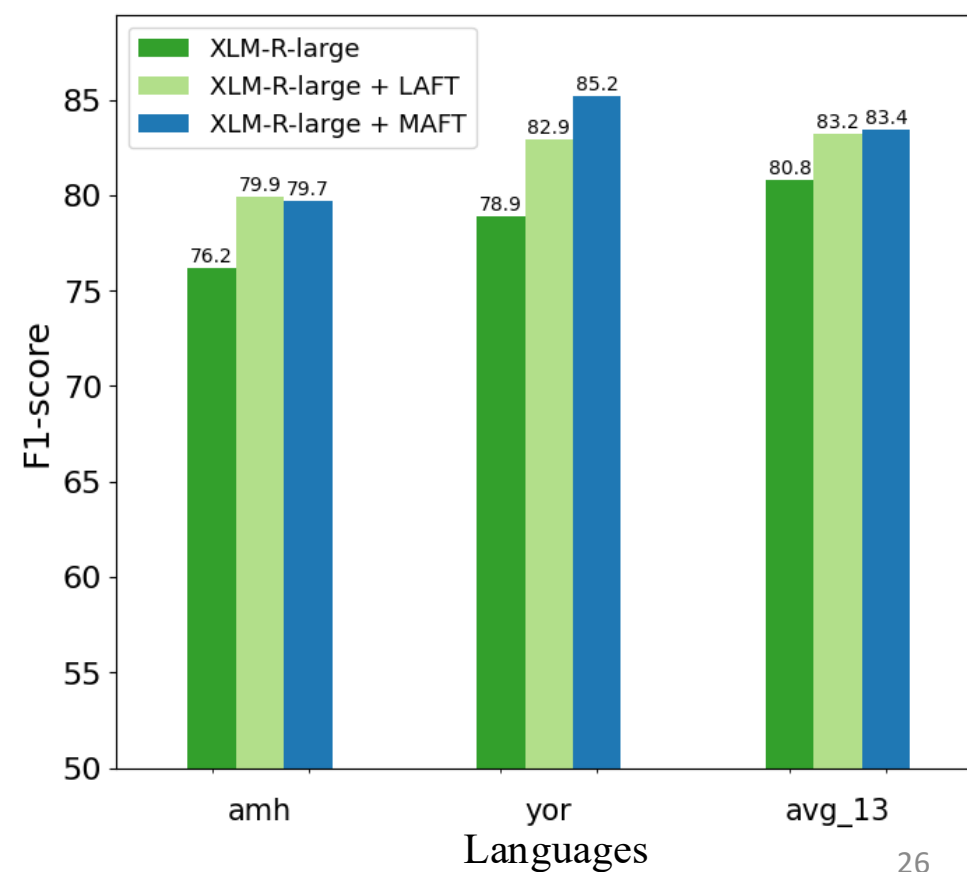


MAFT on MasakhaNER with larger models

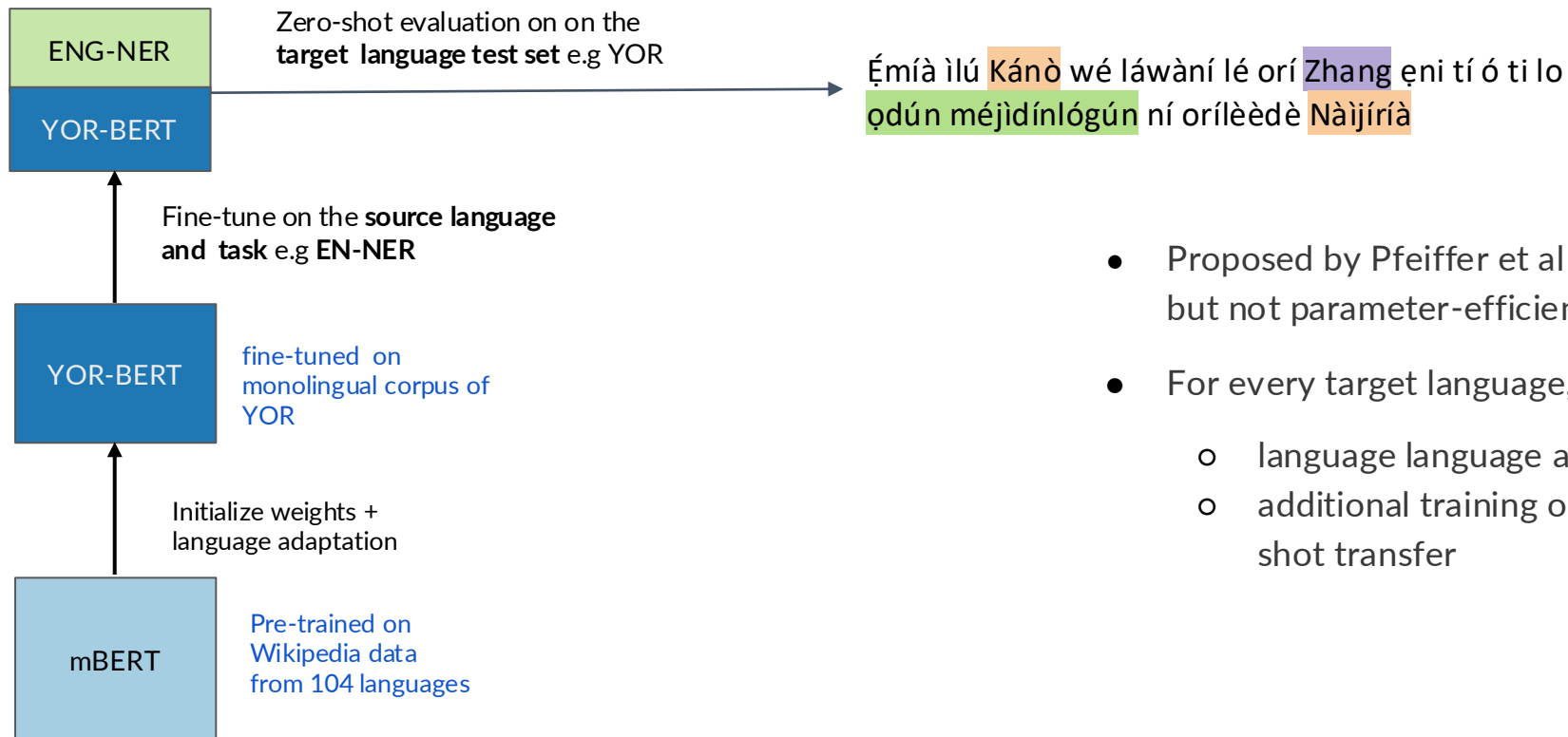
MAFT is either competitive or better than LAFT



scale



Challenge 4: Non-parameter efficient but effective Cross-lingual transfer

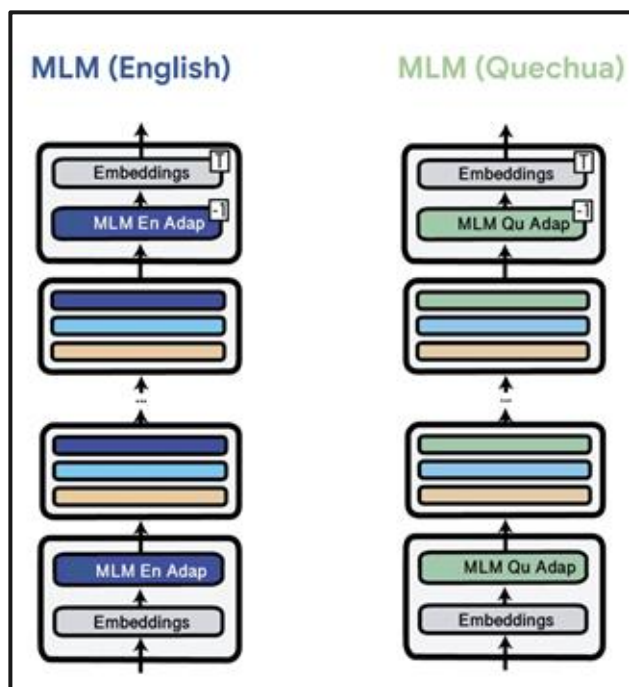


- Proposed by Pfeiffer et al (2020) as **MLM-TRG**, effective but not parameter-efficient.
- For every target language, perform both
 - language language adaptation on monolingual data
 - additional training on source language before zero-shot transfer

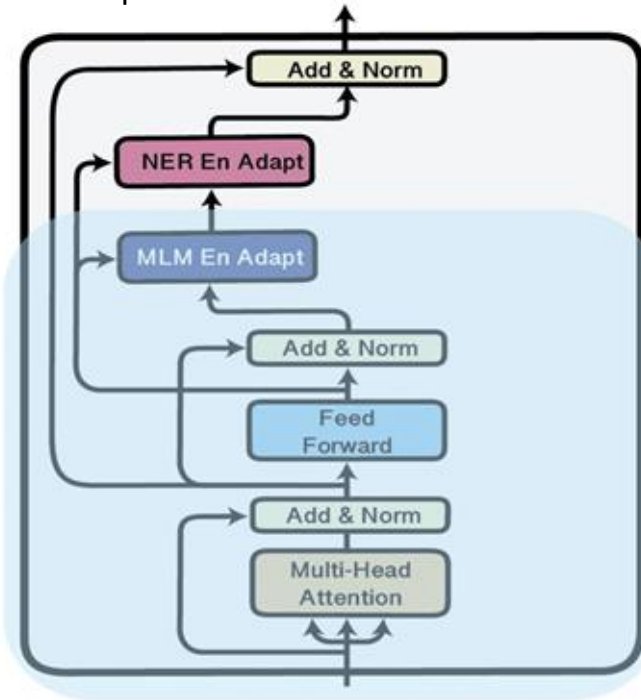
Leveraging target language monolingual data with PEFT and best transfer language

Parameter-efficient fine-tuning with adapters: MAD-X approach

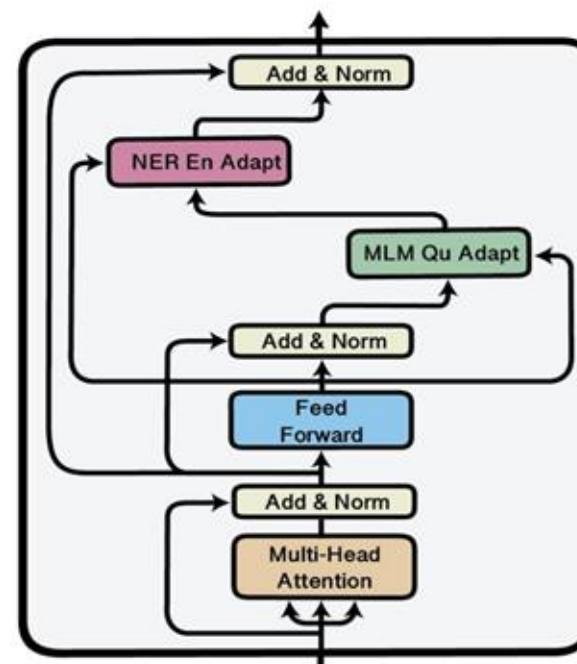
STEP 1: Train language adapters on every language of interest



STEP 2: Train task adapter together with the language adapter, only modify task adapter, others parameters are frozen

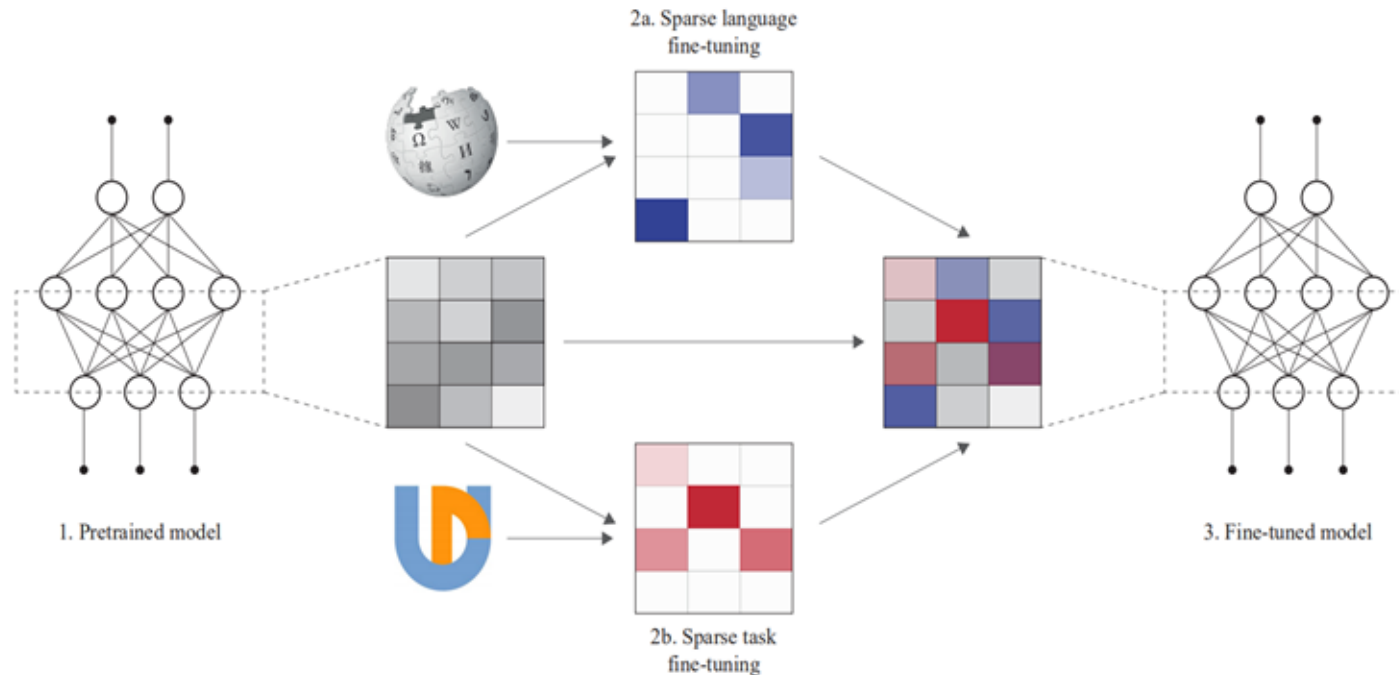


STEP 3: Zero-Shot Transfer to Target Language
by replacing the source language adapter but keeping the task adapter.



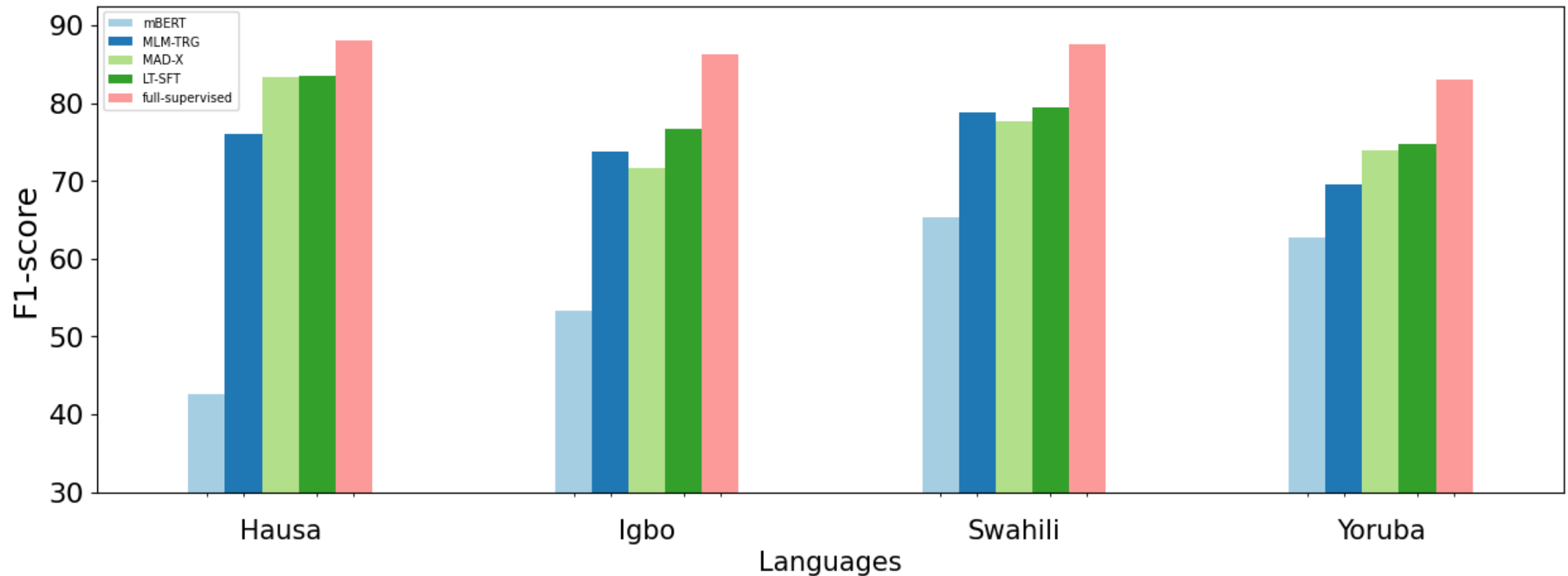
Lottery Ticket Sparse Fine-tunings (LT-SFT)

Lottery Ticket Hypothesis (LTH) (Frankle & Carbin, 2019) that states that **each neural model contains a sub-network (a "winning ticket")** that, *if trained again in isolation, can reach or even surpass the performance of the original model*.



Cross-lingual transfer results on NER with PEFT

Evaluation on MasakhaNER – named entity recognition for African languages.



More effective PEFT Cross-lingual transfer

Leverage a stronger multilingual encoder created by **MAFT** and in-domain data

Evaluation on MasakhaNER

Model	amh	hau	ibo	kin	lug	luo	pcm	swa	wol	yor	avg
XLM-R-base (fully-supervised)	69.7	91.0	86.2	73.8	80.5	75.8	86.9	88.7	69.6	78.1	81.2
mBERT (MAD-X) (Ansell et al., 2021)	-	83.4	71.7	65.3	67.0	52.2	72.1	77.6	65.6	74.0	69.9
mBERT (MAD-X on news domain)	-	86.0	77.6	69.9	73.3	56.9	78.5	80.2	68.8	75.6	74.1
XLM-R-base (MAD-X on news domain)	47.5	85.5	83.2	72.0	75.7	57.8	76.8	84.0	68.2	72.2	75.0
AfroXLMR-base (MAD-X on news domain)	47.7	88.1	80.9	73.0	80.1	59.2	79.9	<u>86.9</u>	69.1	75.6	77.0
mBERT (LT-SFT) (Ansell et al., 2021)	-	83.5	76.7	67.4	67.9	54.7	74.6	79.4	66.3	74.8	71.7
mBERT (LT-SFT on news domain)	-	86.4	80.6	69.2	76.8	55.1	80.4	82.3	71.6	76.7	75.4
XLM-R-base (LT-SFT on news domain)	<u>54.1</u>	87.6	81.4	72.7	79.5	<u>60.7</u>	81.2	85.5	73.6	73.7	77.3
AfroXLMR-base (LT-SFT on news domain)	54.0	<u>88.6</u>	<u>83.5</u>	73.8	81.0	<u>60.7</u>	<u>81.7</u>	86.4	74.5	78.7	<u>78.8</u>

Challenge 5: Choosing the best source language

The **best source language** to transfer from is **not always English**

To further improve performance, you should choose the best transfer

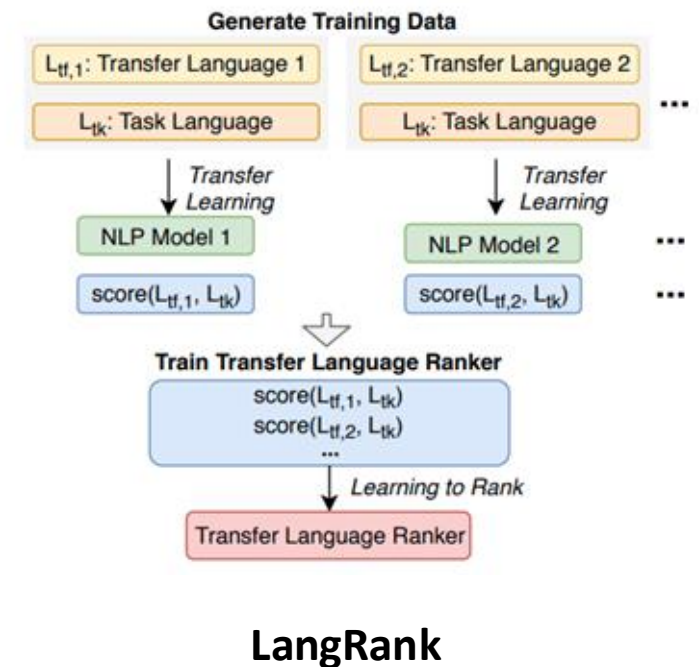
- **Linguistic distance measures from lang2vec**

- Geographic distance (d_{geo})
- Genetic distance (d_{gen})
- Inventory distance (d_{inv})
- Syntactic distance (d_{syn})
- Phonological distance (d_{pho})
- Featural distance (d_{fea})

- **Data-dependent features:**

- Transfer language dataset size (s_{tf})
- Target language dataset size (s_{tg})
- Transfer over target size ratio (sr)
- Entity Overlap (eo)

- **Source-target transfer evaluation score**



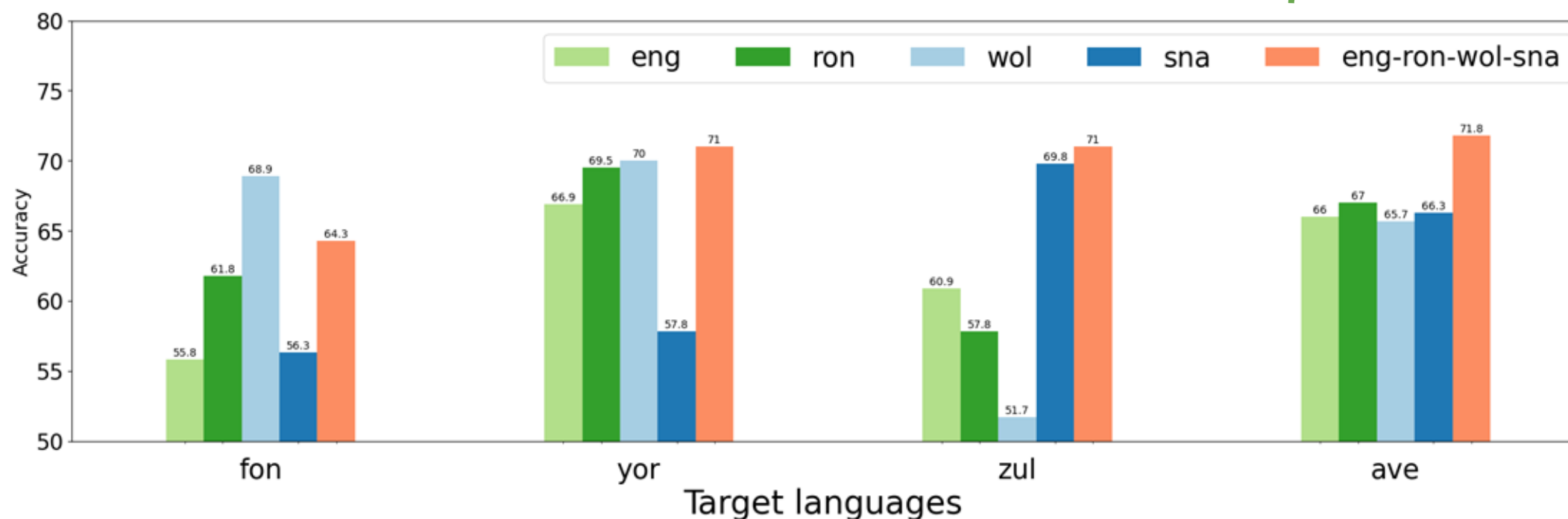
Effect of choosing a better source language: POS

How important is the **source language** using MAD-X adaptation?

- English, Romanian, Wolof, and Shona

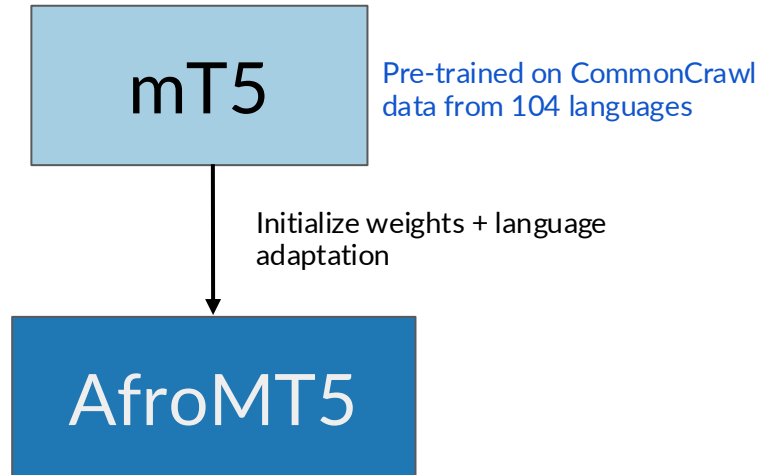
Evaluation on: **Fon**, **Yoruba**, and **isiZulu**

Multi-source/co-training adaptation is very effective



LLM adaptation

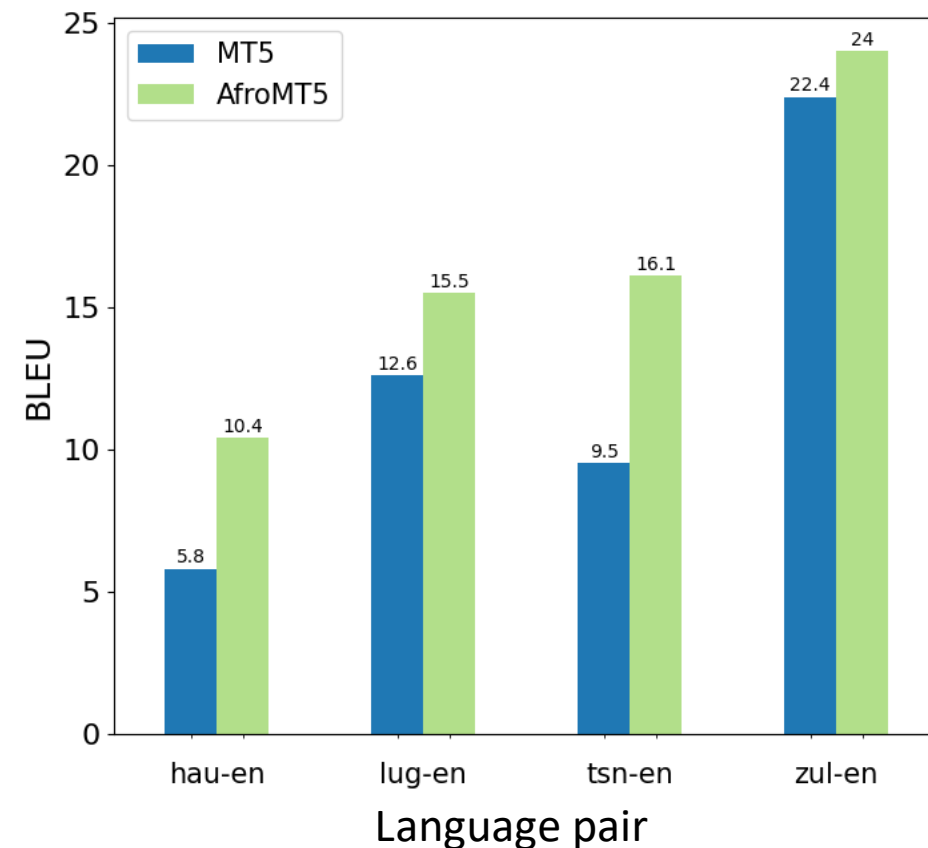
MAFT for Sequence-to-Sequence Models



A Few Thousand Translations Go A Long Way! Leveraging Pre-trained Models for African News Translation

David Ifeoluwa Adelani^{1*}, Jesujoba Oluwadara Alabi^{2*}, Angela Fan^{3*}, Julia Kreutzer^{4*}, Xiaoyu Shen⁵, Machel Reid^{6*}, Dana Ruiter¹, Dietrich Klakow¹, Peter Nabende^{7*}, Ernie Chang^{1*}, Tajuddeen R. Gwadabe^{8*}, Freshia Sackey^{9*}, Bonaventure F. P. Dossou^{10*}, Chris Chinenye Emezue^{11*}, Colin Leong^{12*}, Michael Beukman^{13*}, Shamsuddeen H. Muhammad^{14*}, Guyo D. Jarso^{*}, Oreen Yousuf^{15*}, Andre N. Rubungo^{16*}, Gilles Hacheme^{17*}, Eric P. Wairagala^{7*}, Muhammad U. Nasir^{18*}, Benjamin A. Ajibade^{*}, Tunde Oluwaseyi Ajayi^{*}, Yvonne Wambui Gitau^{*}, Jade Abbott^{*}, Mohamed Ahmed^{19*}, Millicent Ochieng^{19*}, Anuoluwapo Aremu^{*}, Perez Ogayo^{20*}, Jonathan Mukiibi^{7*}, Fatoumata Ouoba Kabore^{*}, Godson Koffi Kalipe^{*}, Derguene Mbaye^{21*}, Allahsera Auguste Tapo^{22*}, Victoire M. Koagne^{*}, Edwin Munkoh-Buabeng^{*}, Valencia Wagner^{23*}, Idris Abdulmumin^{24*}, Ayodele Awokoya^{25*}, Happy Buzaaba^{*}, Blessing Sibanda^{26*}, Andiswa Bukula^{27*}, Sam Manthalu²⁸

Leveraging **pre-trained models** using a few thousand parallel sentences to build a machine translation model for a **new domain**



Language adaptation for Large Language Models

Continue pre-training (CPT) on monolingual data + **Supervised Fine-tuning (SFT)**

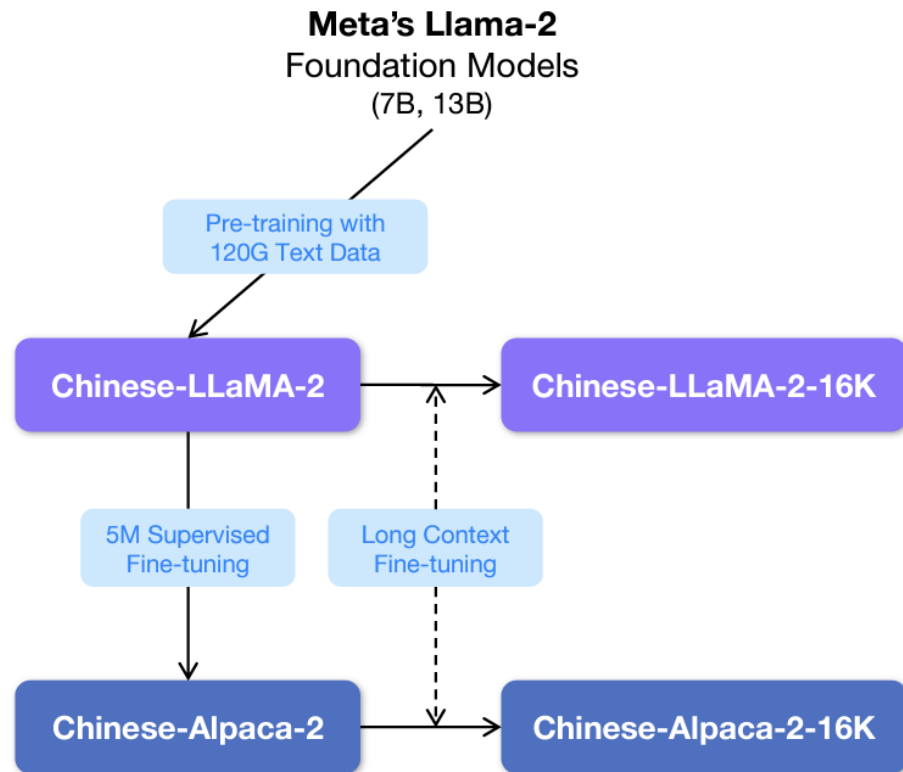


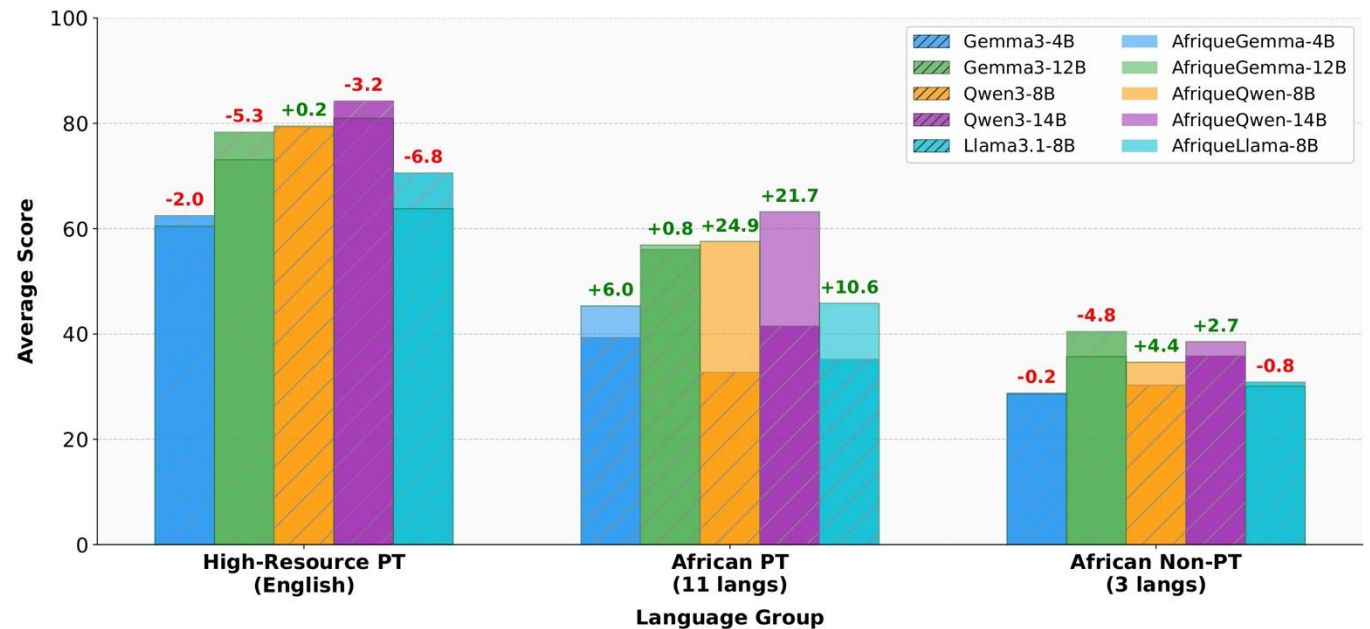
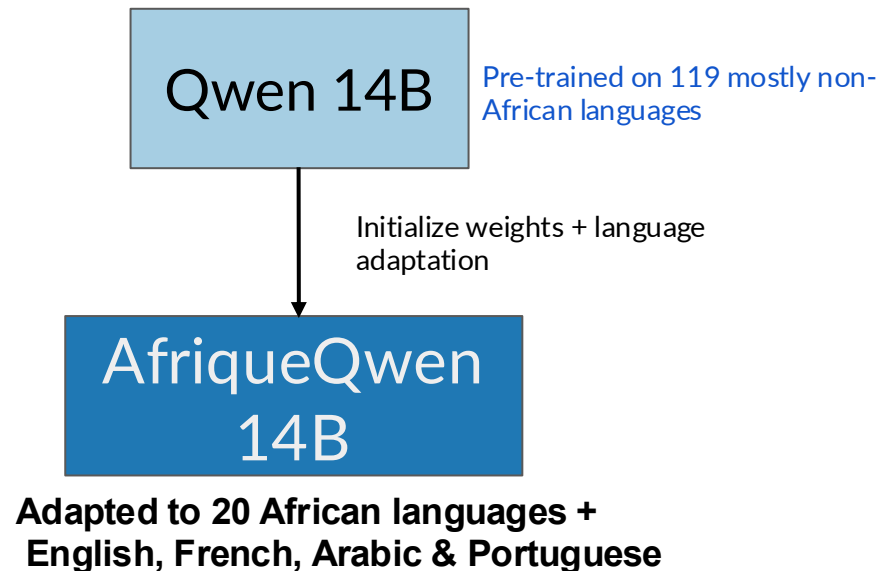
Table 11: **Results on CMMLU test sets.**

Model	Test Set	
	Zero-shot	Few-shot
Chinese-LLaMA-2-7B	27.9	34.1
Chinese-LLaMA-2-13B	38.9	42.5
Chinese-Alpaca-2-7B	40.0	41.8
Chinese-Alpaca-2-13B	43.2	45.5

How effective is CPT on low-resource languages?

Continue-pre-training (CPT) on monolingual data **is not enough**. We need to diversify the data to

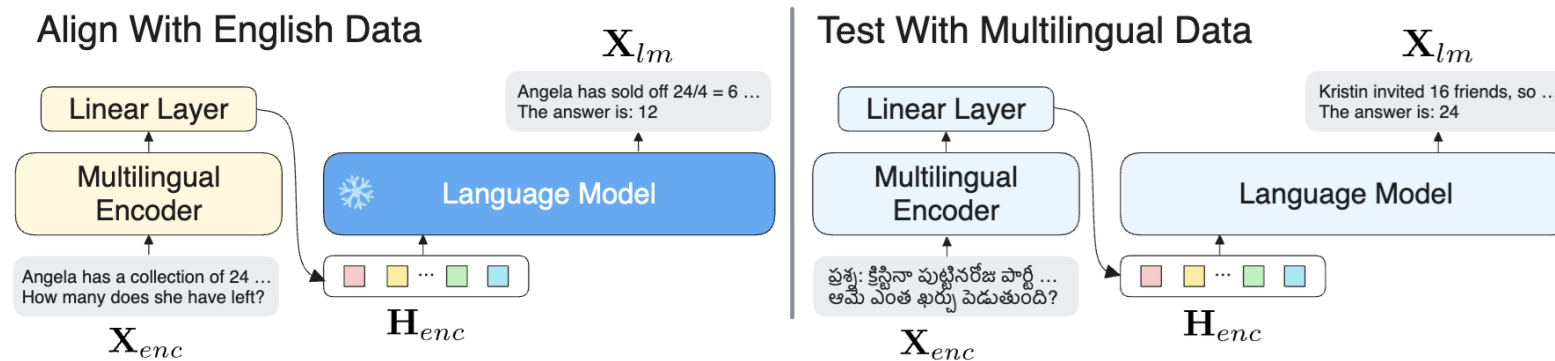
- Cover **more domains** e.g. health, history, industrial, politics, sci & tech, software, travel,
- Add **data with structured reasoning** e.g. **Math** and **Code** data



Evaluation on **AfroBench-Lite**

LLM adaptation with Model stacking

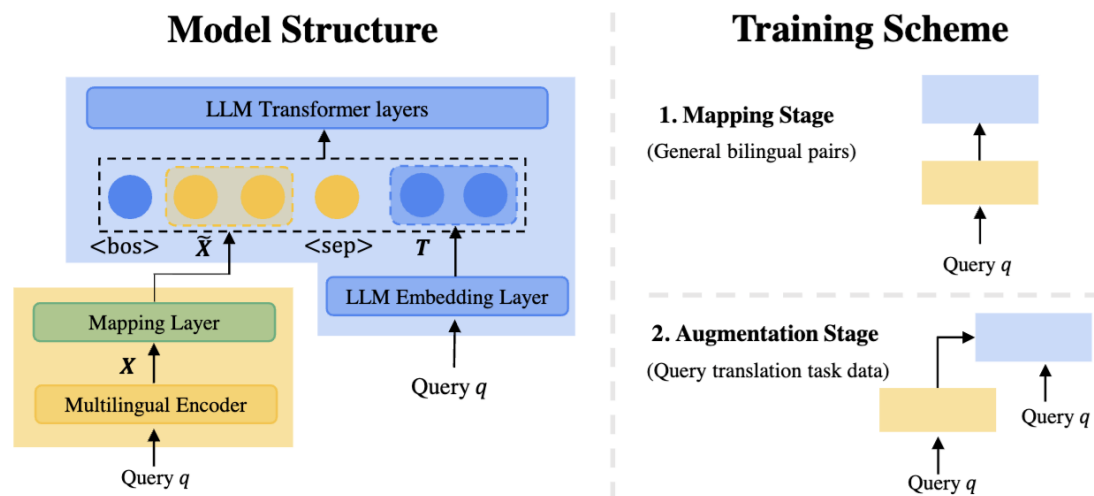
- Earlier LLMs often **do not cover** many low-resource languages, often few dozens
- But we have strong encoders covering many languages e.g. Glot500.
- Can we bridge them?
 - Learn an **adapter** to connect them.
- Introducing **LangBridge**



LLM adaptation with Model stacking

MindMerger: Similar to LangBridge, it follows two steps:

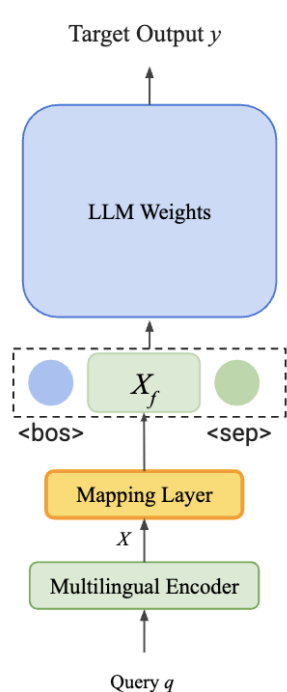
1. **Mapping Stage**: Use translation data from a **general domain** to align an encoder model with a decoder LLM
2. **Augmentation Stage**: Use **task-specific data** generated from a public translation model: English $\rightarrow$ XX



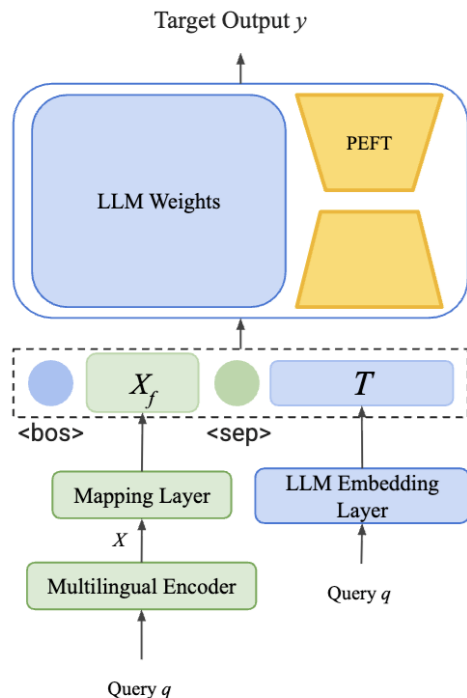
LLM adaptation with Model stacking

MERLIN: Adds two more steps to MindMerger incorporating Curriculum learning:

I - Model Stacking Stage



II - Task Specialization Stage



I - Model Stacking Stage Datasets

I.a Mapping

Tout homme qui connaîtra
cette voie, cette vérité,
cette vie, possède toutes
les bénédictions de Dieu.

Everyone who knows this
Way, this Truth, and this
Life, will have all of God's
blessings!

I.b Question Alignment

Pour quelle valeur de k la
droite représentée par
l'équation $1-kx = -3y$
contient le point $(4,-3)$?

For what value of k does
the line represented by the
equation $1-kx = -3y$ contain
the point $(4,-3)$?

I.c Augmentation

Quelle est la mesure du
degré du complément du
complément d'un angle de
42 degrés?

The complement of a
42-degree angle is $90^\circ - 42^\circ = 48^\circ$. The supplement
of a 48-degree angle is
 $180^\circ - 48^\circ = 132^\circ$. The
answer is: 132

II - Task Specialization Stage Dataset

LLM Task Specialization

Quelle est la mesure du
degré du complément du
complément d'un angle de
42 degrés?

The complement of a
42-degree angle is $90^\circ - 42^\circ = 48^\circ$. The supplement
of a 48-degree angle is
 $180^\circ - 48^\circ = 132^\circ$. The
answer is: 132

MERLIN vs. other Model stacking methods

MERLIN exceeds the performance of MindMerger and LangBridge

Model	Bn	Th	Sw	Ja	Zh	De	Fr	Ru	Es	En	MRL	HRL	Avg
MGSM													
<i>Open Models</i>													
Metamath 7B	6.4	2.0	4.4	36.4	40.4	52.8	50.4	46.0	58.4	63.6	4.3	49.7	36.1
Gemma 2 9B	56.4	66.4	64.4	65.2	69.6	70.4	59.2	67.6	72.0	76.0	62.4	68.6	66.7
<i>Key Multilingual Approaches–Metamath 7B & mT5-XL</i>													
LangBridge	42.8	50.4	43.2	40.0	45.2	56.4	50.8	52.4	58.0	63.2	45.5	52.3	50.2
QAlign-LoRA	32.4	39.6	40.4	44.0	48.4	54.8	56.8	52.4	59.6	68.0	37.5	54.9	49.6
SLAM	32.0	44.0	40.0	46.0	48.4	54.0	55.2	54.8	56.8	64.8	38.7	54.3	49.6
LayAlign	1.6	3.6	8.0	23.2	23.2	37.2	37.6	34.8	38.4	43.2	4.4	33.9	25.1
MindMerger	43.6	44.4	42.5	45.6	45.6	56.8	58.0	56.0	54.0	64.0	43.5	54.3	51.1
MERLIN	43.2	49.2	44.8	48.4	48.8	53.2	55.6	58.0	59.2	63.6	45.7	55.3	52.4
Δ (MERLIN-MM)	−0.4	+4.8	+2.3	+2.8	+3.2	−3.6	−2.4	+2.0	+5.2	−0.4	+2.2	+1.0	+1.3
<i>Key Multilingual Approaches–Gemma 2 9B & NLLB-600M-distilled</i>													
QAlign-LoRA	58.0	60.8	62.0	56.4	62.4	68.8	68.0	68.0	76.8	80.8	60.3	68.7	66.2
SLAM	63.2	61.2	52.4	62.4	64.4	68.8	53.6	70.0	69.2	69.2	58.9	65.4	63.4
MindMerger	66.8	72.8	69.2	66.8	72.8	78.0	75.2	73.6	76.4	82.4	69.6	75.0	73.4
MERLIN	74.8	76.0	77.6	71.6	72.8	73.2	74.8	78.4	77.6	85.6	76.1	76.3	76.2
Δ (MERLIN-MM)	+8.0	+3.2	+8.4	+4.8	0.0	−4.8	−0.4	+4.8	+1.2	+3.2	+6.5	+1.3	+2.8

Summary

- We described various challenges to cross-lingual transfer and how to address them
 - **Lack of benchmark data:** create one
 - **Limited language coverage in LM:** several effective adaptation techniques exist.
 - **Adapting to new scripts:** perform better embedding initialization
 - **Parameter inefficiency of some adaptation techniques:** use more parameter efficient methods such as MAD-X
 - **Choice of the best source language:** Use LangRank or related methods.
 - Approaches are still **applicable in the LLM age**.
 - With some slight modification